AYMANE AIT DADS

AI & ML Intern | Model Experimentation · Data Preparation · Performance Evaluation

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PROFESSIONAL SUMMARY

Ranked 17th out of 3,000+ teams in the NVIDIA Nemotron Reasoning Challenge (score 0.86/1.0, $106K+ prize pool) by fine-tuning a 30B-parameter model with LoRA SFT - directly demonstrating the model experimentation and performance evaluation skills central to this role. Reduced manual network analysis time by 60% at Orange Maroc by building Random Forest and K-Means models on 100,000+ records, delivering actionable maintenance recommendations to stakeholders. Brings hands-on experience in data preparation, ablation studies, PyTorch training pipelines, and cross-domain ML experimentation across computer vision, NLP, and large language model fine-tuning. Aims to apply staged adapter continuation training, task-level diagnostic methodology, and rigorous model evaluation practices to real-world business challenges at Resolve Tech Solutions.

CORE COMPETENCIES

Model Experimentation | Data Preparation | Performance Evaluation | Machine Learning Models | Artificial Intelligence | Model Development | Data Science | Business Challenges

WORK EXPERIENCE

Orange Maroc
Jul 2024 - Aug 2024

Data Science Intern - Casablanca, Morocco

- Built **Random Forest and K-Means machine learning models** on 100,000+ fiber-optic network records - classifying network quality across thousands of nodes using upload speed, latency, and jitter features.
- Developed a clustering pipeline mapping customer performance profiles to **5 commercial service tiers**, with output adopted directly by the network optimization team for infrastructure decisions.
- Reduced manual analysis time by **~60%** through automated feature engineering and model evaluation workflows, accelerating the team's ability to act on network performance data.
- Delivered stakeholder presentations translating model outputs into **concrete maintenance recommendations**, bridging technical findings and business decision-making using Power BI dashboards.

PROJECTS

NVIDIA Nemotron Model Reasoning Challenge — *Kaggle Competition · In Progress (June 2026) · Ranked 17th / 3,000+ Teams*

- Ranked **17th / 3,000+ teams** (score 0.86/1.0, $106K+ prize pool) in the ongoing NVIDIA Nemotron Reasoning Challenge by fine-tuning a **30B-parameter Mamba-Transformer** with LoRA SFT (r=32, BF16, completion-only loss, 8,192-token context).
- Performed **task-level error analysis** across 5 reasoning categories (unit conversion, cipher, symbolic, gravity, bit manipulation); built a diagnostic evaluation pipeline revealing that reasoning over-expansion - not incorrect reasoning - was the primary failure mode.

PyTorch · Unsloth · TRL · PEFT · Hugging Face Transformers · vLLM · Kaggle GPU

AI-Generated Image Detection — *EURECOM ImSecu + NTIRE 2026 @ CVPR · 2025 · Ranked 1st in Class*

- Ranked **1st in EURECOM class leaderboard** (0.791 AUC) by fine-tuning CLIP ViT-L/14 (428M parameters) with a custom MLP head on 250,000 images spanning 25+ generator types; also submitted independently to the NTIRE 2026 @ CVPR CodaBench international benchmark.
- Applied **10-view test-time augmentation** and a 3-model ensemble weighted by validation AUC, discovering that 1-epoch training (0.793 AUC) outperformed 4-epoch training (0.767 AUC) due to superior out-of-distribution generalization.

PyTorch · OpenCLIP · ViT · FP16 Mixed Precision · AdamW

EDUCATION

Engineering Degree in Data Science (Master-level) - EURECOM — *Sep 2023 - Present*

Relevant coursework: Machine Learning, Deep Learning, Statistics, Computer Vision, NLP, Cloud Computing, Image Security, Web Applications

CPGE - Mathematics & Physics - Ibn Timiya — *2021 - 2023*

SKILLS

LLM Fine-Tuning & Reasoning: LoRA · QLoRA · SFTTrainer · Completion-Only Loss · Chain-of-Thought · vLLM · Mamba-Transformer

ML / DL & Data Science: PyTorch · Scikit-learn · Hugging Face Transformers · PEFT · Pandas · NumPy · Ensemble Methods

MLOps, CV & NLP: CLIP · ViT · FP16/BF16 Mixed Precision · Ablation Studies · Feature Engineering · Sentiment Analysis · Power BI